

Game Theory & Strategy

Comprehensive Week-by-Week Notes

IITM BS Degree Programme

Exam Preparation Guide

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Contents

1	Week 1: Basics of Game Theory	2
1.1	Preferences and Rationality	2
1.2	Components of a Game	2
1.3	Nash Equilibrium	2
1.4	Optimization in Games	3
2	Week 2: Strategic Form Games	4
2.1	Dominant Strategies	4
2.2	Prisoner's Dilemma	4
2.3	Iterated Elimination of Dominated Strategies (IEDS)	4
2.4	Mixed Strategies	4
2.5	Best Response Functions	5
3	Week 3: Extensive Form Games	6
3.1	Game Trees	6
3.2	Backward Induction	6
3.3	Subgame Perfect Equilibrium	6
4	Week 4: Auctions and Mechanism Design	7
4.1	Auction Types	7
4.2	Vickrey (Second-Price) Auction	7
4.3	First-Price Sealed-Bid Auction	7
4.4	Expected Revenue	8
4.5	Revenue Equivalence Theorem	8
4.6	Order Statistics	8
5	Week 5: Cooperative Game Theory — Basics	9
5.1	Coalitional Games	9
5.2	The Core	9
5.3	Properties	9
6	Week 6: Social Choice and Voting	10
6.1	Voting Mechanisms	10
6.2	Information Cascades	10

7	Week 7: Matching Theory	11
7.1	Stable Matching	11
7.2	Gale-Shapley Algorithm (Deferred Acceptance)	11
8	Week 8: Evolutionary Game Theory & Repeated Games	12
8.1	Evolutionary Stable Strategy (ESS)	12
8.2	Hawk-Dove Game	12
8.3	Repeated Games	12
9	Week 9: Network Effects & Market Design	13
9.1	Network Goods — Standard Model	13
9.2	Variant: $U(\theta) = \theta(a + 3\theta vn^e)$	13
10	Week 10: Cooperative Game Theory — Weighted Majority & Shapley	15
10.1	Weighted Majority Games	15
10.2	Shapley-Shubik Power Index	16
10.3	Shapley Value — General Games	16
11	Week 11: Resource Division & Cost Allocation	17
11.1	Contested Garment Principle	17
11.2	Cost Allocation in Networks (Cable TV)	17
12	Week 12: VCG Mechanism & Ad Auctions	19
12.1	VCG (Vickrey-Clarke-Groves) Mechanism	19
13	Additional Key Topics	20
13.1	Cournot Competition (Duopoly)	20
13.2	Market Entry Game	20
13.3	Declining Industry (Exit Game)	21
13.4	Roommate Cleaning Game	21
13.5	Bayes' Theorem (Defective Items)	21
14	Exam Focus Topics & Strategy	23

1 Week 1: Basics of Game Theory

1.1 Preferences and Rationality

Rational Preferences

A preference relation $\succsim$ over a set of alternatives is **rational** if it satisfies:

1. **Completeness:** For any two alternatives A, B : either $A \succsim B$ or $B \succsim A$ (or both).
2. **Reflexivity:** For any alternative A : $A \succsim A$.
3. **Transitivity:** If $A \succsim B$ and $B \succsim C$, then $A \succsim C$.

Violation of Transitivity

Suppose a person prefers Tea $\succ$ Coffee, Coffee $\succ$ Lassi, but Lassi $\succ$ Tea. This violates transitivity and leads to irrational cycling — a “money pump” can exploit such preferences.

1.2 Components of a Game

Normal Form Game

A normal form (strategic form) game $G = (N, S, u)$ consists of:

- $N = \{1, 2, \dots, n\}$: set of **players**
- S_i : set of **strategies** available to player i ; $S = S_1 \times S_2 \times \dots \times S_n$
- $u_i : S \rightarrow \mathbb{R}$: **payoff function** for player i

1.3 Nash Equilibrium

Nash Equilibrium (NE)

A strategy profile $s^* = (s_1^*, s_2^*, \dots, s_n^*)$ is a Nash Equilibrium if for every player i :

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall s_i \in S_i$$

No player can improve their payoff by unilaterally deviating.

Key Insight

At a Nash Equilibrium, every player is playing a **best response** to the strategies of all other players.

1.4 Optimization in Games

Resource Allocation

A student has H hours total to study for two quizzes. If she allocates h hours to Quiz 1 and $(H - h)$ to Quiz 2, with score functions $f_1(h)$ and $f_2(H - h)$, she maximizes:

$$\max_{0 \leq h \leq H} [f_1(h) + f_2(H - h)]$$

FOC: $f'_1(h^*) = f'_2(H - h^*)$ (equalize marginal returns).

2 Week 2: Strategic Form Games

2.1 Dominant Strategies

Strictly Dominant Strategy

Strategy s_i **strictly dominates** s'_i if:

$$u_i(s_i, s_{-i}) > u_i(s'_i, s_{-i}) \quad \forall s_{-i} \in S_{-i}$$

Weakly Dominant Strategy

Strategy s_i **weakly dominates** s'_i if:

$$u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i}) \quad \forall s_{-i}$$

with strict inequality for at least one s_{-i} .

2.2 Prisoner's Dilemma

Prisoner's Dilemma Payoff Matrix

	C	D
C	(-1, -1)	(-3, 0)
D	(0, -3)	(-2, -2)

Defect (D) is a strictly dominant strategy for both players.

NE: (D, D) with payoff (-2, -2), which is Pareto-inferior to (C, C) at (-1, -1).

2.3 Iterated Elimination of Dominated Strategies (IEDS)

IEDS Procedure

1. Identify and remove all strictly dominated strategies for each player.
2. In the reduced game, repeat step 1.
3. Continue until no more strategies can be eliminated.
4. If a unique strategy profile remains, it is the NE.

Note: Order of elimination does not matter for strictly dominated strategies, but *does* matter for weakly dominated strategies.

2.4 Mixed Strategies

Mixed Strategy

A mixed strategy for player i is a probability distribution σ_i over S_i . The expected payoff is:

$$U_i(\sigma_1, \sigma_2) = \sum_{s_1 \in S_1} \sum_{s_2 \in S_2} \sigma_1(s_1) \cdot \sigma_2(s_2) \cdot u_i(s_1, s_2)$$

Finding Mixed Strategy NE (2-player, 2-strategy)

If Player 1 plays U with probability p and D with probability $(1 - p)$, at the MSNE Player 2 must be **indifferent** between their strategies:

$$E[u_2(L)] = E[u_2(R)]$$

Solve for p . Similarly, Player 1's indifference gives Player 2's mixing probability q .

Matching Pennies

	H	T
H	$(1, -1)$	$(-1, 1)$
T	$(-1, 1)$	$(1, -1)$

No pure strategy NE. MSNE: both play H and T with probability $\frac{1}{2}$ each.

2.5 Best Response Functions

Best Response

Player i 's best response to s_{-i} :

$$BR_i(s_{-i}) = \arg \max_{s_i \in S_i} u_i(s_i, s_{-i})$$

Nash Equilibrium: $s_i^* \in BR_i(s_{-i}^*)$ for all i .

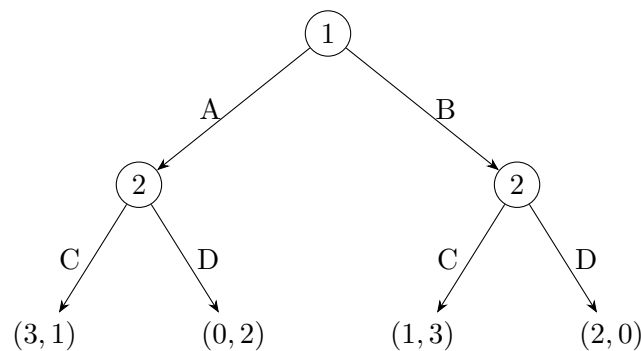
3 Week 3: Extensive Form Games

3.1 Game Trees

Extensive Form Game

An extensive form game specifies:

- The set of players
- The order of moves (game tree)
- Information sets (what each player knows when moving)
- Actions at each decision node
- Payoffs at terminal nodes



3.2 Backward Induction

Backward Induction Algorithm

1. Start at **terminal nodes** and identify optimal choices at the last decision nodes.
2. Replace those decision nodes with the resulting payoffs.
3. Repeat, moving backward up the tree until the root is reached.
4. The resulting strategy profile is the **Subgame Perfect Nash Equilibrium (SPNE)**.

Zermelo's Theorem

Every finite game of perfect information has a pure strategy NE that can be found by backward induction. In particular, games like chess have a determined outcome (though we may not know it).

3.3 Subgame Perfect Equilibrium

Subgame Perfect Nash Equilibrium

A strategy profile is a SPNE if it constitutes a Nash Equilibrium in **every subgame** of the original game. SPNE eliminates non-credible threats.

4 Week 4: Auctions and Mechanism Design

4.1 Auction Types

Standard Auction Formats

- **English Auction:** Ascending price; last bidder wins. Dominant strategy: bid up to your valuation.
- **Dutch Auction:** Descending price; first bidder to accept wins.
- **First-Price Sealed-Bid (FPSB):** Highest bidder wins, pays their bid.
- **Second-Price Sealed-Bid (Vickrey):** Highest bidder wins, pays the *second-highest* bid.
- **All-Pay Auction:** Everyone pays their bid, only highest bidder wins.

4.2 Vickrey (Second-Price) Auction

Dominant Strategy in Vickrey Auction

In a second-price sealed-bid auction, bidding your true valuation $b_i = v_i$ is a **weakly dominant strategy**.

4.3 First-Price Sealed-Bid Auction

Symmetric Equilibrium Bidding Strategy — FPSB

With n bidders and valuations i.i.d. Uniform $[0, 1]$:

$$b(v) = \frac{n-1}{n} \cdot v$$

Each bidder “shades” their bid below their valuation.

FPSB Computation

5 bidders, valuation $v = 0.6$:

$$b(0.6) = \frac{5-1}{5} \times 0.6 = \frac{4}{5} \times 0.6 = 0.48$$

6 bidders, valuation $v = 0.9$:

$$b(0.9) = \frac{6-1}{6} \times 0.9 = \frac{5}{6} \times 0.9 = 0.75$$

4.4 Expected Revenue

Expected Revenue Computations

- $n = 5$: $E[R] = \frac{4}{6} = \frac{2}{3}$
- $n = 6$: $E[R] = \frac{5}{7}$

4.5 Revenue Equivalence Theorem

Revenue Equivalence

If bidders are risk-neutral, have independent private valuations drawn from the same distribution, and the bidder with the highest valuation wins, then all standard auction formats yield the same expected revenue.

4.6 Order Statistics

Expected Value of k -th Order Statistic

If $X_1, \dots, X_n \sim \text{Uniform}[0, 1]$ i.i.d., and $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$:

$$E[X_{(k)}] = \frac{k}{n+1}$$

Order Statistic Examples from Exams

- 5 numbers, 2nd position: $E[X_{(2)}] = \frac{2}{6} = \frac{1}{3}$
- 6 numbers, 3rd position: $E[X_{(3)}] = \frac{3}{7}$

5 Week 5: Cooperative Game Theory — Basics

5.1 Coalitional Games

Coalitional (TU) Game

A coalitional game (N, v) consists of:

- $N = \{1, 2, \dots, n\}$: set of players
- $v : 2^N \rightarrow \mathbb{R}$: **characteristic function** (coalition function) with $v(\emptyset) = 0$
- $v(S)$ represents the total payoff coalition S can guarantee for itself

5.2 The Core

Core

The **core** of a game (N, v) is the set of feasible allocations $(x_1, \dots, x_n)$ such that:

1. **Efficiency**: $\sum_{i \in N} x_i = v(N)$
2. **Coalitional rationality**: $\sum_{i \in S} x_i \geq v(S) \quad \forall S \subseteq N$

5.3 Properties

- **Individual Rationality**: $x_i \geq v(\{i\})$ for all i
- The core may be empty, a single point, or a convex polytope
- **Superadditivity**: $v(S \cup T) \geq v(S) + v(T)$ for disjoint S, T

6 Week 6: Social Choice and Voting

6.1 Voting Mechanisms

Condorcet Winner

An alternative x is a **Condorcet winner** if it beats every other alternative in pairwise majority voting.

Agenda Voting / Sequential Elimination

Alternatives are matched pairwise in a specified order (agenda). The winner of each round faces the next alternative. The order of the agenda can determine the final winner — this is a form of **agenda manipulation**.

6.2 Information Cascades

Information Cascade

An information cascade occurs when individuals, acting sequentially, rationally choose to **ignore their own private signal** and follow the actions of predecessors. This can lead to herd behavior, where the crowd converges on a potentially wrong action.

Key Exam Points on Crowds

- “Crowd can never be wrong even if everyone is rational” — **FALSE**
- “Aggregate behaviour of many with limited info can never produce accurate results” — **FALSE**
- Both statements are false: crowds can be wrong (cascades), and crowds can also be surprisingly accurate (wisdom of crowds / Condorcet jury theorem).

7 Week 7: Matching Theory

7.1 Stable Matching

Stable Matching

A matching μ is **stable** if there is no **blocking pair**: a pair (m, w) not matched to each other such that both prefer each other over their current partners.

Important: A stable matching does *not* mean everyone is satisfied — it only means no unmatched pair can profitably deviate together.

7.2 Gale-Shapley Algorithm (Deferred Acceptance)

Gale-Shapley Algorithm

1. Each man proposes to his most preferred woman (who hasn't rejected him yet).
2. Each woman with multiple proposals **tentatively holds** the best one and rejects the rest.
3. Rejected men propose to their next choice.
4. Repeat until no rejections occur.

Properties:

- Always terminates and produces a stable matching.
- **Men-proposing** $\Rightarrow$ **men-optimal** (best stable partner for every man) and **women-pessimal**.
- Women-proposing gives the opposite.

8 Week 8: Evolutionary Game Theory & Repeated Games

8.1 Evolutionary Stable Strategy (ESS)

ESS

A strategy s^* is an **Evolutionary Stable Strategy** if for any mutant strategy $s \neq s^*$:

1. $u(s^*, s^*) > u(s, s^*)$, **or**
2. $u(s^*, s^*) = u(s, s^*)$ and $u(s^*, s) > u(s, s)$

8.2 Hawk-Dove Game

Hawk-Dove

	H	D
H	$(\frac{V-C}{2}, \frac{V-C}{2})$	$(V, 0)$
D	$(0, V)$	$(\frac{V}{2}, \frac{V}{2})$

If $V < C$: mixed ESS where fraction $p^* = V/C$ plays Hawk.

If $V > C$: Hawk is ESS.

8.3 Repeated Games

Infinitely Repeated Games

When a stage game is repeated infinitely with discount factor $\delta \in (0, 1)$:

$$\text{Total payoff} = \sum_{t=0}^{\infty} \delta^t \cdot u_t$$

Grim Trigger Strategy

- Cooperate in round 1.
- Continue cooperating as long as the opponent cooperated in all previous rounds.
- If opponent ever defects, defect forever.

Cooperation sustained if: $\delta \geq \frac{\text{temptation} - \text{cooperation}}{\text{temptation} - \text{punishment}}$

9 Week 9: Network Effects & Market Design

9.1 Network Goods — Standard Model

Network Good Utility

Consumer of type θ (uniformly distributed on $[0, 1]$) gets utility:

$$U(\theta) = \theta(a + vn^e) - p$$

where a = standalone benefit, $v > 0$ = network effect, n^e = expected network size, p = price.

Key Derivations — $U(\theta) = \theta(a + vn^e)$

Indifferent consumer: $U(\hat{\theta}) = p$ gives:

$$\hat{\theta} = \frac{p}{a + vn^e}$$

Maximum price for buyers: $\hat{\theta} \leq 1$ requires:

$$p \leq a + vn^e$$

Mass of buyers: Consumers with $\theta \geq \hat{\theta}$ buy:

$$n = 1 - \hat{\theta}$$

Willingness to pay (inverse demand): Setting $n^e = n$ in equilibrium:

$$p(n^e, n) = (a + vn^e)(1 - n)$$

Fulfilled-expectations demand curve ($n^e = n$):

$$p = (a + vn)(1 - n)$$

Properties of $p(n^e, n)$

- **Decreasing** function of n (more sold $\rightarrow$ lower marginal WTP)
- **Increasing** function of n^e (larger expected network $\rightarrow$ higher WTP)

9.2 Variant: $U(\theta) = \theta(a + 3\theta vn^e)$

This variant appeared in the Dec 2025 exam. Here consumers with higher θ also value the network more.

Derivations for $U(\theta) = a\theta + 3\theta^2 vn^e$

Set $U(\hat{\theta}) = p$:

$$a\hat{\theta} + 3vn^e\hat{\theta}^2 = p \implies \hat{\theta} = \frac{p - a}{3vn^e} \quad (\text{simplified for MCQ form: } \hat{\theta} = \frac{p - a}{3vn^e})$$

(The exact form depends on the question variant; check the quadratic carefully.)

Max price: $p \leq a + 3vn^e$

Mass of buyers: $n = 1 - \hat{\theta}$ gives $n = \frac{a-p+3vn^e}{3vn^e}$

WTP: $p(n^e, n) = a + 3vn^e(1 - n)$

Fulfilled-expectations demand (with $m = 1 - n$): $p = a + 3vm(1 + m)$

10 Week 10: Cooperative Game Theory — Weighted Majority & Shapley

10.1 Weighted Majority Games

Weighted Majority Game

A weighted majority game $[q; w_1, w_2, \dots, w_n]$ consists of:

- q : quota (votes needed to pass)
- w_i : weight (votes) of player i
- $v(S) = 1$ if $\sum_{i \in S} w_i \geq q$, else $v(S) = 0$

Symmetric and Null Players

- **Symmetric players:** i and j are symmetric if $v(S \cup \{i\}) = v(S \cup \{j\})$ for every $S \subseteq N \setminus \{i, j\}$.
- **Null player:** Player i is null if $v(S \cup \{i\}) = v(S)$ for every $S \subseteq N$.

Game $[61; 40, 40, 30, 10]$

— — — *FullCoalitionFunction*]Quota $q = 61$. *Weights* : $w_1 = 40, w_2 = 40, w_3 = 30, w_4 = 10$.

Coalition S	$\sum w_i$	$v(S)$
$\{1\}$	40	0
$\{2\}$	40	0
$\{3\}$	30	0
$\{4\}$	10	0
$\{1, 2\}$	80	1
$\{1, 3\}$	70	1
$\{1, 4\}$	50	0
$\{2, 3\}$	70	1
$\{2, 4\}$	50	0
$\{3, 4\}$	40	0
$\{1, 2, 3\}$	110	1
$\{1, 2, 4\}$	90	1
$\{1, 3, 4\}$	80	1
$\{2, 3, 4\}$	80	1
$\{1, 2, 3, 4\}$	120	1

Symmetric players: Players 1, 2, 3 are all symmetric (3 symmetric players).

Check: Interchanging any two among $\{1, 2, 3\}$ preserves coalition values. Player 4 is **null** (adding player 4 never changes $v(S)$: e.g., $v(\{1, 4\}) = 0 = v(\{1\})$, $v(\{1, 2, 4\}) = 1 = v(\{1, 2\})$, etc.).

10.2 Shapley-Shubik Power Index

Shapley Value

For a game (N, v) with $|N| = n$:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)]$$

The **Shapley-Shubik power index** is the Shapley value applied to weighted majority games (where $v(S) \in \{0, 1\}$). It equals the fraction of orderings where player i is **pivotal** (changes the coalition from losing to winning).

Shapley-Shubik for $[61; 40, 40, 30, 10]$

]With 4 players, there are $4! = 24$ orderings. Count pivotal positions :

Players 1, 2, 3 are each pivotal in 8 orderings. Player 4 is pivotal in 0.

$$\phi_1 = \phi_2 = \phi_3 = \frac{8}{24} = \frac{1}{3}, \quad \phi_4 = 0$$

So $\frac{1}{x_i} = 3$ for $i = 1, 2, 3$ and $x_4 = 0$.

Null Player in $[7; 5, 1, 3, 4]$

]Quota = 7. Weights : $w_1 = 5, w_2 = 1, w_3 = 3, w_4 = 4$.

- $\{1, 3\} = 8 \geq 7$ (wins); removing player 2 doesn't matter.
- $\{1, 4\} = 9 \geq 7$; $\{3, 4\} = 7 \geq 7$; $\{1, 3, 4\} = 12$.
- Adding player 2 to any coalition: $\{1, 2\} = 6 < 7$, $\{2, 3\} = 4 < 7$, $\{2, 4\} = 5 < 7$.
- Player 2 never makes a losing coalition winning $\Rightarrow$ **Player 2 is null**.

10.3 Shapley Value — General Games

Shapley Value Computation

$N = \{1, 2, 3\}$, $v(1) = v(2) = 5$, $v(3) = 0$, $v(\emptyset) = 0$, $v(1, 2) = 10$, $v(1, 3) = v(2, 3) = 5$, $v(1, 2, 3) = 10$.

Ordering	mc_1	mc_2	mc_3
1-2-3	5	5	0
1-3-2	5	5	0
2-1-3	5	5	0
2-3-1	5	5	0
3-1-2	5	5	0
3-2-1	5	5	0
Sum	30	30	0

$\phi_1 = 30/6 = 5$, $\phi_2 = 30/6 = 5$, $\phi_3 = 0/6 = 0$.

11 Week 11: Resource Division & Cost Allocation

11.1 Contested Garment Principle

Contested Garment Principle

Two creditors with claims c_1 and c_2 divide an estate E . Each creditor “concedes” the portion they don’t claim:

- Creditor 1 concedes: $\max(E - c_1, 0)$ to Creditor 2
- Creditor 2 concedes: $\max(E - c_2, 0)$ to Creditor 1

The **contested amount** = $E - (\text{concession}_1 + \text{concession}_2)$ is split equally.

Contested Garment Computation

$$x_1 = \max(E - c_2, 0) + \frac{E - \max(E - c_1, 0) - \max(E - c_2, 0)}{2}$$

$$x_2 = E - x_1$$

Estate = 500, Claims = 400, 300

- Creditor 1 concedes to 2: $\max(500 - 400, 0) = 100$
- Creditor 2 concedes to 1: $\max(500 - 300, 0) = 200$
- Contested: $500 - 100 - 200 = 200$, split equally: 100 each
- $x_1 = 200 + 100 = 300$, $x_2 = 100 + 100 = 200$

Estate = 500, Claims = 450, 250

- Creditor 1 concedes: $\max(500 - 450, 0) = 50$
- Creditor 2 concedes: $\max(500 - 250, 0) = 250$
- Contested: $500 - 50 - 250 = 200$, split equally: 100 each
- $x_1 = 250 + 100 = 350$, $x_2 = 50 + 100 = 150$

11.2 Cost Allocation in Networks (Cable TV)

Bird’s Rule for Tree Networks

For a tree network connecting players to a source (station):

1. Each player’s **stand-alone cost** = cost of the unique path from station to that player.
2. For each edge in the tree, the cost is shared equally among all players whose path uses that edge.
3. Player i ’s cost = $\sum_{\text{edges } e \text{ on path to } i} \frac{c(e)}{|\text{players using } e|}$

3-Player Cable TV Network

Tree: Station $\rightarrow$ (18) $\rightarrow$ Junction $\rightarrow$ (12) $\rightarrow$ Player 2, Junction $\rightarrow$ (24) $\rightarrow$ Player 1, Junction $\rightarrow$ (12) $\rightarrow$ Player 3.

Edge 18 (Station to Junction): used by all 3 $\Rightarrow$ each pays $18/3 = 6$.

Edge 24 (Junction to P1): only P1 $\Rightarrow$ P1 pays 24.

Edge 12 (Junction to P2): only P2 $\Rightarrow$ P2 pays 12.

Edge 12 (Junction to P3): only P3 $\Rightarrow$ P3 pays 12.

Totals: $P1 = 6 + 24 = 30$...

Wait — the answer key says $P1=6$, $P2=30$, $P3=30$. The tree structure from the image is:

- Station $\rightarrow$ (18) $\rightarrow$ Player 1 $\rightarrow$ (24) $\rightarrow$ Junction $\rightarrow$ (12) $\rightarrow$ Player 2, Junction $\rightarrow$ (12) $\rightarrow$ Player 3

Edge 18 (Station to P1): used by all 3 $\Rightarrow$ each pays $18/3 = 6$.

Edge 24 (P1 to Junction): used by P2 and P3 $\Rightarrow$ each pays $24/2 = 12$.

Edge 12 (Junction to P2): only P2 $\Rightarrow$ P2 pays 12.

Edge 12 (Junction to P3): only P3 $\Rightarrow$ P3 pays 12.

Totals: $P1 = 6$, $P2 = 6 + 12 + 12 = 30$, $P3 = 6 + 12 + 12 = 30$.

12 Week 12: VCG Mechanism & Ad Auctions

12.1 VCG (Vickrey-Clarke-Groves) Mechanism

VCG Mechanism

The VCG mechanism:

1. Allocates to **maximize total social welfare** (sum of valuations).
2. Each winner pays the **externality** they impose on others: the decrease in others' total welfare caused by their presence.

VCG Price for Ad Slots

Given slots with click-through rates (CTR) and advertisers with per-click valuations:

1. **Social optimum:** Assign highest-value advertiser to highest-CTR slot, etc.
2. **VCG price for winner of slot j :**

$$p_j = (\text{Total welfare of others without player } j) - (\text{Welfare of others with player } j)$$

Ad Auction: 2 slots, 3 advertisers

Slots: a (CTR=10), b (CTR=5).

Advertisers: x (value=3/click), y (value=2/click), z (value=1/click).

Social optimum: $x \rightarrow a$, $y \rightarrow b$. Total value = $3 \times 10 + 2 \times 5 = 40$.

VCG price for slot a (advertiser x):

- Without x : assign $y \rightarrow a$, $z \rightarrow b$. Others' welfare = $2 \times 10 + 1 \times 5 = 25$.
- With x : others' welfare = y gets slot $b = 2 \times 5 = 10$.
- VCG price for $a = 25 - 10 = 15$.

VCG price for slot b (advertiser y):

- Without y : assign $x \rightarrow a$, $z \rightarrow b$. Others' welfare = $3 \times 10 + 1 \times 5 = 35$.
- With y : others' welfare = x gets slot $a = 3 \times 10 = 30$.
- VCG price for $b = 35 - 30 = 5$.

13 Additional Key Topics

13.1 Cournot Competition (Duopoly)

Cournot Model

Two firms, zero marginal cost, demand $p = a - Q$ where $Q = q_1 + q_2$.

Profit of firm i : $\pi_i = q_i(a - q_1 - q_2)$

Best response: $\frac{\partial \pi_i}{\partial q_i} = 0 \implies q_i = \frac{a - q_j}{2}$

Nash Equilibrium: $q_1^* = q_2^* = \frac{a}{3}$

Equilibrium price: $p^* = \frac{a}{3}$

Equilibrium profit: $\pi_i^* = \left(\frac{a}{3}\right)^2 = \frac{a^2}{9}$

Cournot with $p = 200 - Q$

$a = 200$, zero costs.

$$q_1^* = q_2^* = \frac{200}{3} \approx 66.67$$

$$\pi_1^* = \pi_2^* = \left(\frac{200}{3}\right)^2 = \frac{40000}{9} \approx 4444$$

$$\text{Best response if } q_1 = 100: q_2 = \frac{200 - 100}{2} = 50$$

13.2 Market Entry Game

3 Firms, Payoff = $150/n - 62$

Payoff from entering when n firms enter: $150/n$. Cost of entering: 62.

Net payoff from entering when n enter: $150/n - 62$.

- $n = 1$: $150 - 62 = 88 > 0$ (profitable)
- $n = 2$: $75 - 62 = 13 > 0$ (profitable)
- $n = 3$: $50 - 62 = -12 < 0$ (not profitable)

Pure NE: Exactly 2 firms enter. There are $\binom{3}{2} = 3$ such equilibria.

“All three enter” is *not* a NE $\implies$ **FALSE**.

Mixed NE: By symmetry, each enters with probability p . A firm is indifferent when:

$$E[\text{payoff from Enter}] = E[\text{payoff from Stay Out} = 0]$$

$$\binom{2}{0}(1-p)^2 \cdot 88 + \binom{2}{1}p(1-p) \cdot 13 + \binom{2}{2}p^2 \cdot (-12) = 0$$

$$88(1-p)^2 + 26p(1-p) - 12p^2 = 0$$

$$\text{Expanding: } 88 - 176p + 88p^2 + 26p - 26p^2 - 12p^2 = 0$$

$$88 - 150p + 50p^2 = 0 \implies 50p^2 - 150p + 88 = 0$$

$$p = \frac{150 \pm \sqrt{22500 - 17600}}{100} = \frac{150 \pm 70}{100}$$

$$p = 2.2 \text{ (reject) or } p = 0.8 = 4/5. \text{ So } \boxed{p = 4/5}.$$

13.3 Declining Industry (Exit Game)

Two Firms, E/T/N Strategies

E = exit immediately, T = exit this quarter, N = exit next quarter.

Payoffs: both operating = -1 per quarter each; one alone = $+2$ per quarter.

	E	T	N
E	(0, 0)	(0, 2)	(0, 4)
T	(2, 0)	(-1, -1)	(-1, 1)
N	(4, 0)	(1, -1)	(-2, -2)

Dominated strategies: No *strictly* dominated strategies (True).

T is *weakly* dominated (by N or E depending on the opponent's choice).

Pure NE: (E, N) and (N, E) — one exits immediately, the other stays both quarters.

Mixed NE: Unique. After eliminating T (weakly dominated), solve on {E, N}:

	E	N
E	(0, 0)	(0, 4)
N	(4, 0)	(-2, -2)

Player 2 indifferent: $0 \cdot \alpha + 4(1 - \alpha) = 0$... Let α = prob of E, β = prob of N. $\alpha + \beta = 1$ (since T eliminated). For Player 2 indifferent: $0 \cdot \alpha + 4 \cdot \beta = 0 \cdot \alpha + (-2)\beta$... Actually let's redo this.

Player 1 mixes E with prob α , N with prob $\beta = 1 - \alpha$. Player 2's payoff from E: $0 \cdot \alpha + 0 \cdot \beta = 0$. Player 2's payoff from N: $4\alpha + (-2)\beta = 4\alpha - 2(1 - \alpha) = 6\alpha - 2$.

Indifference: $0 = 6\alpha - 2 \implies \alpha = 1/3, \beta = 2/3. \beta/\alpha = 2$.

13.4 Roommate Cleaning Game

$$\text{Payoff} = (10 - t_j)t_i - t_i^2$$

Each roommate chooses cleaning time $t_i \geq 0$.

Best response: $\frac{\partial}{\partial t_i}[(10 - t_j)t_i - t_i^2] = (10 - t_j) - 2t_i = 0$

$$BR_i(t_j) = \frac{10 - t_j}{2}$$

If $t_1 = 0$: $BR_2(0) = 5$.

Any $t_i > 5$ is dominated by $t_i = 5$ (since BR_i maxes at 5 when $t_j = 0$).

Symmetric NE: $t_1 = t_2 = t^*$, so $t^* = \frac{10 - t^*}{2} \implies 2t^* = 10 - t^* \implies t^* = \frac{10}{3}$.

Writing as $t_1 = t_2 = \alpha/3$, we get $\alpha = 10$.

13.5 Bayes' Theorem (Defective Items)

Bayes' Theorem Application

Machines A, B, C produce 50%, 30%, 20% of items with defect rates 3%, 4%, 5%.

P(Defective):

$$P(D) = 0.5 \times 0.03 + 0.3 \times 0.04 + 0.2 \times 0.05 = 0.015 + 0.012 + 0.01 = 0.037$$

$P(\text{Machine A} \mid \text{Defective})$:

$$P(A|D) = \frac{P(D|A) \cdot P(A)}{P(D)} = \frac{0.03 \times 0.5}{0.037} = \frac{0.015}{0.037} \approx 0.4054$$

14 Exam Focus Topics & Strategy

High-Priority Topics (Appear in Every Exam)

1. Weighted majority games: coalition function, symmetric/null players
2. Shapley-Shubik power index computation
3. VCG mechanism for ad auctions
4. Network effects: indifferent consumer, demand curves, WTP
5. Contested garment principle
6. First-price auction equilibrium bidding: $b(v) = \frac{n-1}{n}v$
7. Expected revenue: $\frac{n-1}{n+1}$
8. Order statistics: $E[X_{(k)}] = \frac{k}{n+1}$
9. Cournot duopoly: best response, NE quantities, profits
10. Bayes' theorem / defective items

Medium-Priority Topics

1. Cost allocation in tree networks (cable TV)
2. Market entry games with mixed strategies
3. Declining industry / exit games
4. Stable matching (definition, NOT algorithm details)
5. Information cascades (conceptual True/False)
6. Roommate cleaning game (continuous strategies, BR, NE)

Conceptual Questions (Quick Points)

1. "Stable matching $\neq$ everyone satisfied" (Answer: None / D)
2. "Crowd can never be wrong" (FALSE)
3. "Aggregate behaviour can never be accurate" (FALSE)
4. Dominant vs. dominated strategies (strict vs. weak)

